LIVERPOOL JOHN MOORES UNIVERSITY

Predicting Purchase Intention in E-Commerce Using Behavioural Features with Traditional Machine Learning and Autoencoder Models

Soumya Roy

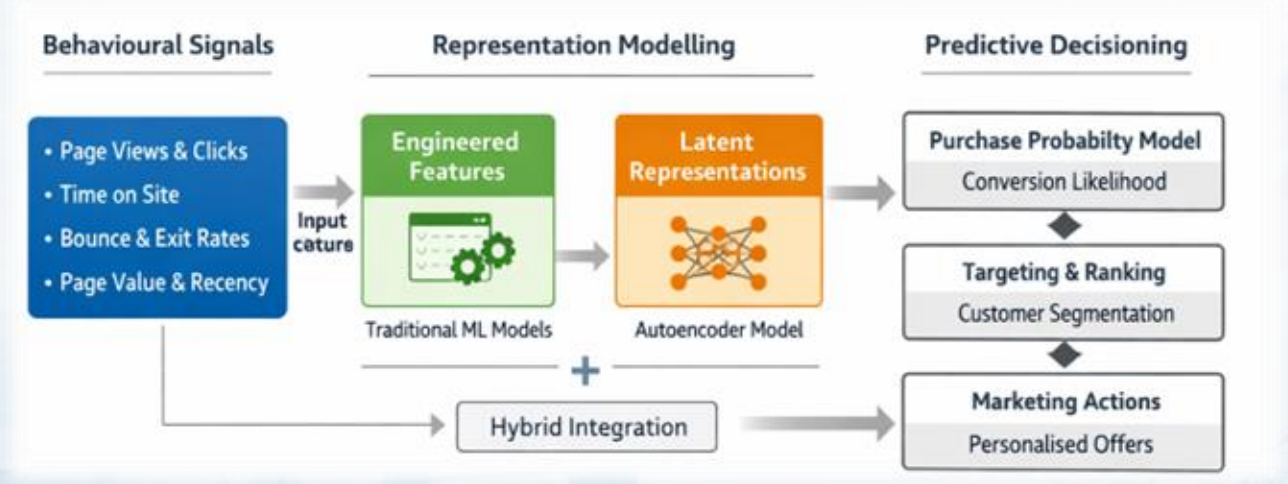
MSc Data Science & Artificial Intelligence

Liverpool John Moores University

March 2026

upGrad

Background



- Rapid expansion of digital commerce
- Session-level behavioural tracking
- 12,330 anonymised sessions
- 15% purchase rate (imbalanced classification)

Research Problem

- Most studies focus on single modelling paradigm
- Limited comparison of engineered vs latent representation
- Robustness rarely tested across seeds
- Calibration & threshold often ignored
- Hybrid integration underexplored

Research Gap

- Limited comparison between engineered behavioural features and latent representations
- Lack of robustness validation across multiple random seeds
- Calibration and threshold optimisation rarely examined
- Hybrid integration between representations underexplored
- Limited interpretation of behavioural misclassification patterns

Aim & Objectives

LIVERPOOL JOHN MOORES UNIVERSITY

Aim

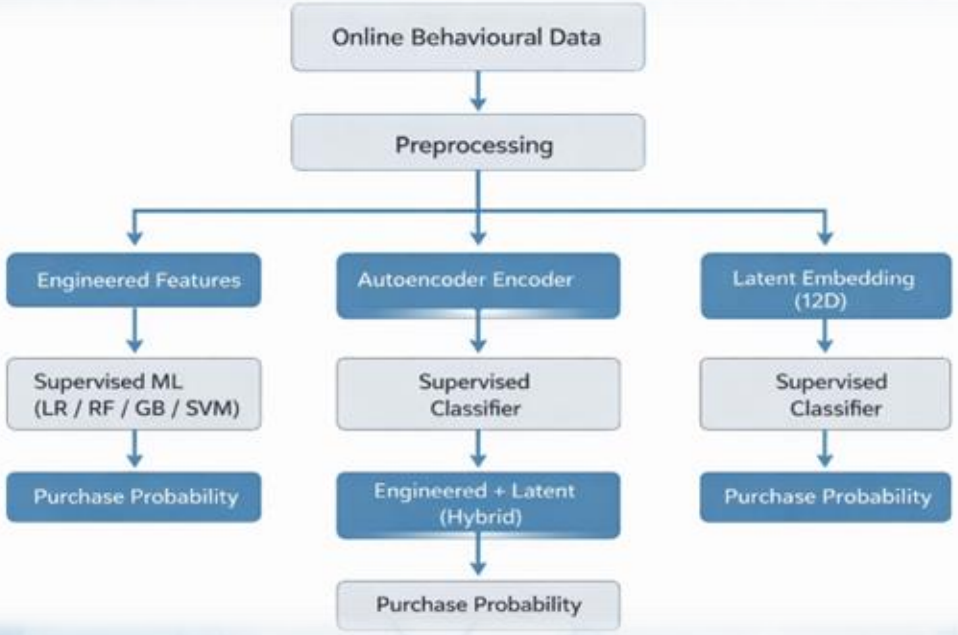
To evaluate and compare engineered behavioural modelling and autoencoder-based representation learning under robust validation.

Objectives

- Develop traditional ML classifiers
- Train autoencoder for latent compression
- Perform multi-seed cross-validation
- Evaluate calibration & threshold optimisation
- Test hybrid integration
- Analyse behavioural misclassification

upGrad

Literature Review



Behavioural Modelling

- Engagement depth correlates with conversion
- Page value signals transactional intent
- Exit and bounce reflect abandonment

Traditional ML

- Logistic Regression
- Random Forest
- Gradient Boosting
- SVM

Literature Review

LIVERPOOL JOHN MOORES UNIVERSITY

Representation Learning

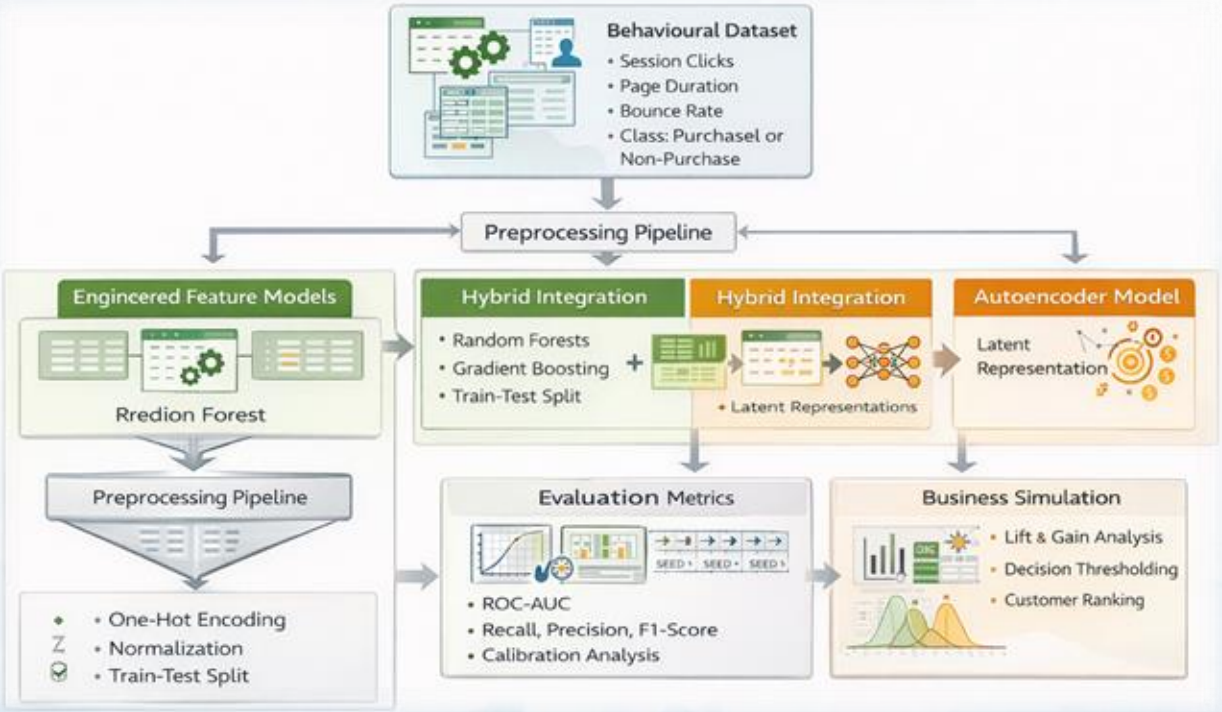
- Autoencoders compress feature space
- Learn latent behavioural structure
- Reduce dimensionality

Research Gap

Does latent compression improve discrimination in structured tabular data?

upGrad

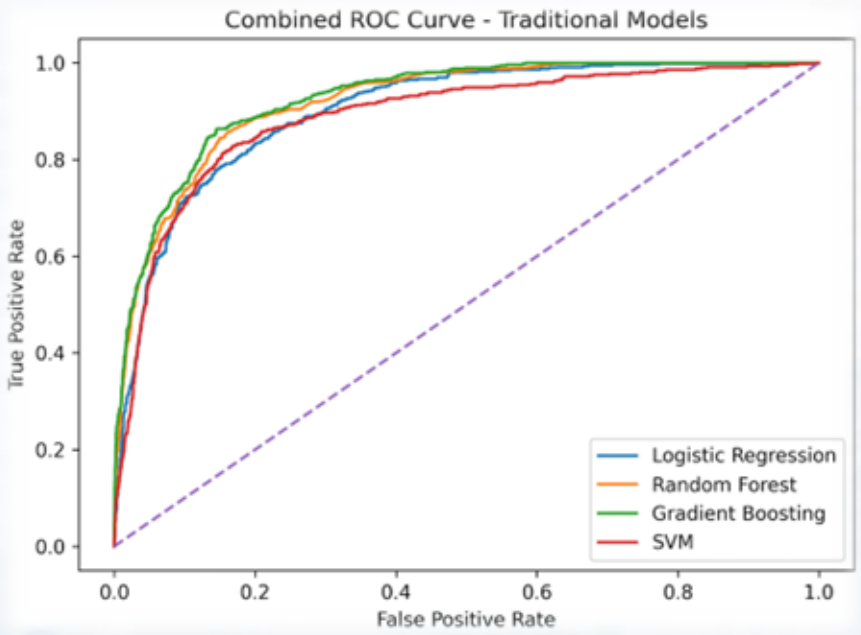
Methodology



Experimental Design

- Stratified 70–30 train-test split
- One-hot encoding + Standard scaling
- 4 Traditional classifiers
- Autoencoder (12 latent dimensions)
- Multi-seed 10-fold cross-validation
- Calibration & threshold analysis

Traditional Model Results



Model	Accuracy	Precision (Buyer)	Recall (Buyer)	F1	ROC-AUC
Logistic Regression	0.86	0.52	0.75	0.62	0.900
Random Forest	0.90	0.75	0.53	0.62	0.918
Gradient Boosting	0.90	0.71	0.58	0.64	0.926
SVM	0.86	0.54	0.75	0.63	0.888

Performance Summary

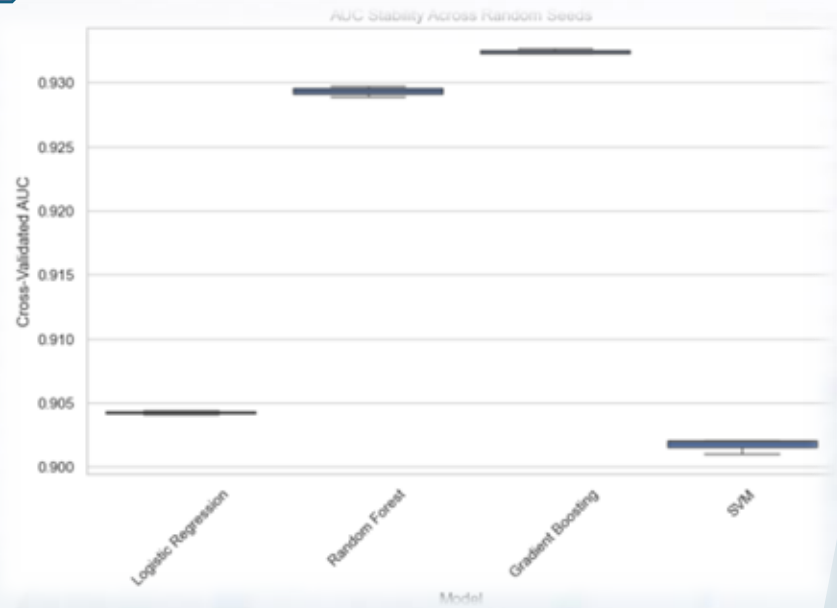
Gradient Boosting

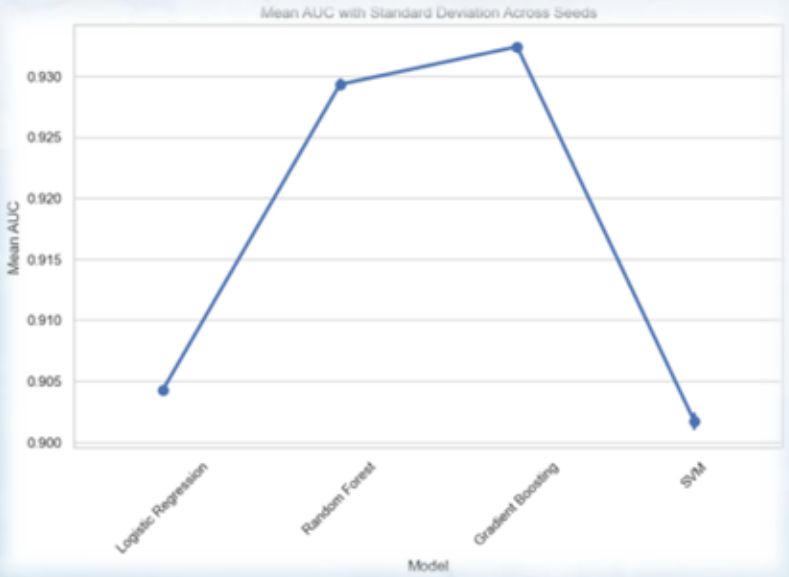
ROC–AUC ≈ 0.93

Strong precision-recall balance

LIVERPOOL JOHN MOORES UNIVERSITY
upGrad

Robustness Analysis

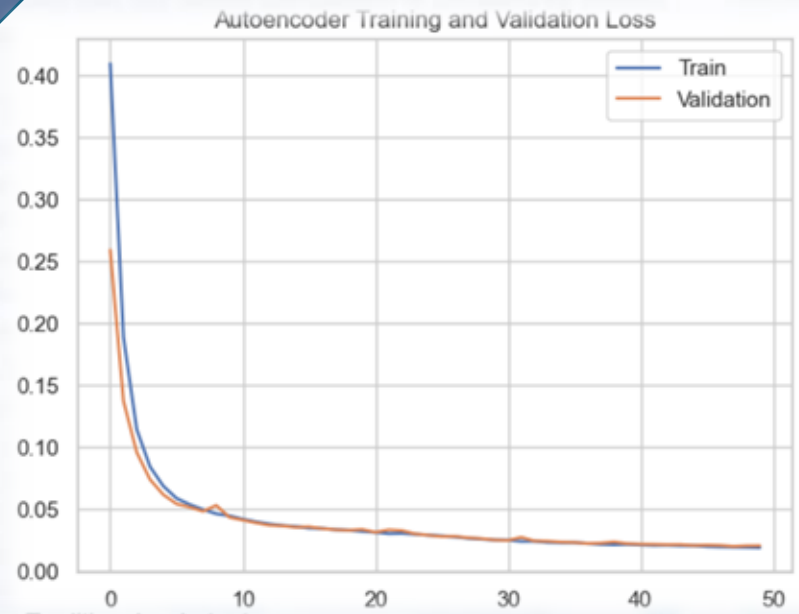


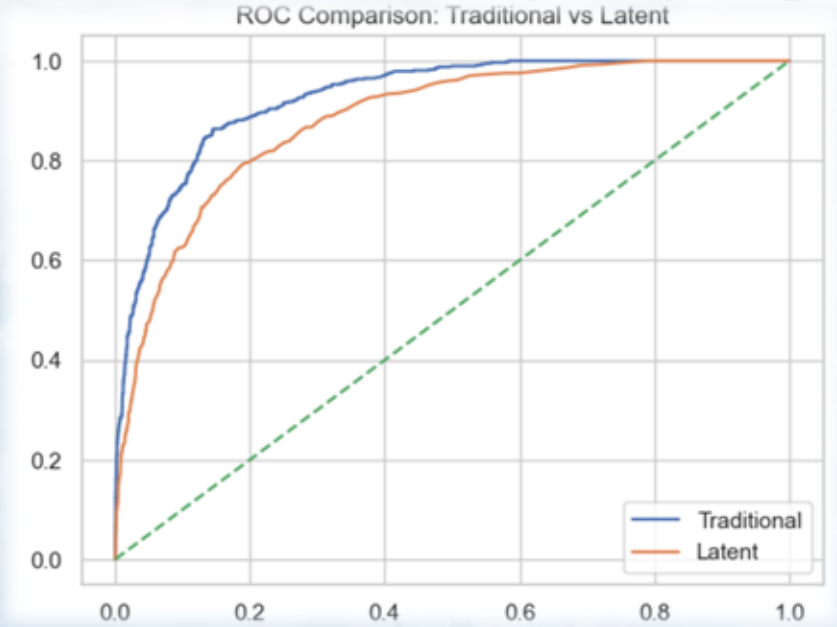


Stability Across Seeds

- Low AUC variance
- Consistent recall
- Minimal fluctuation

Representation Learning Results



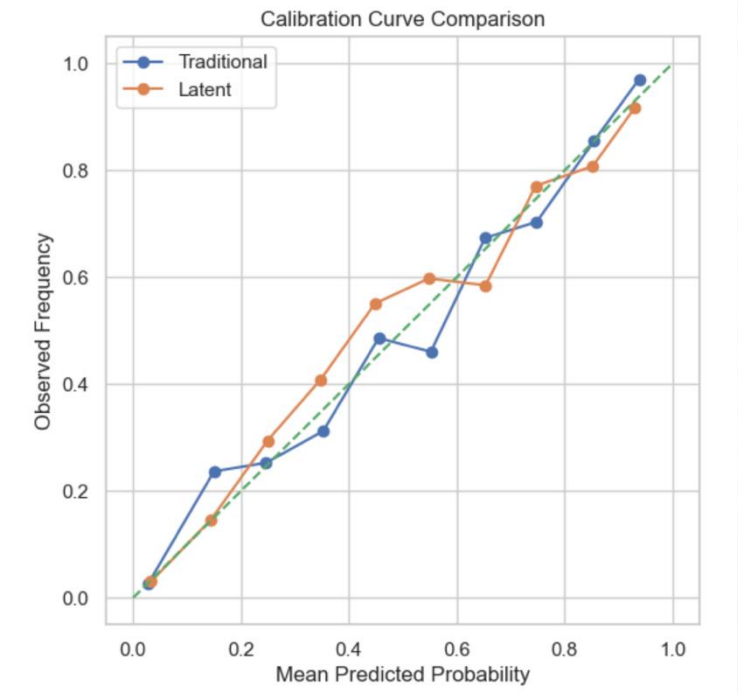


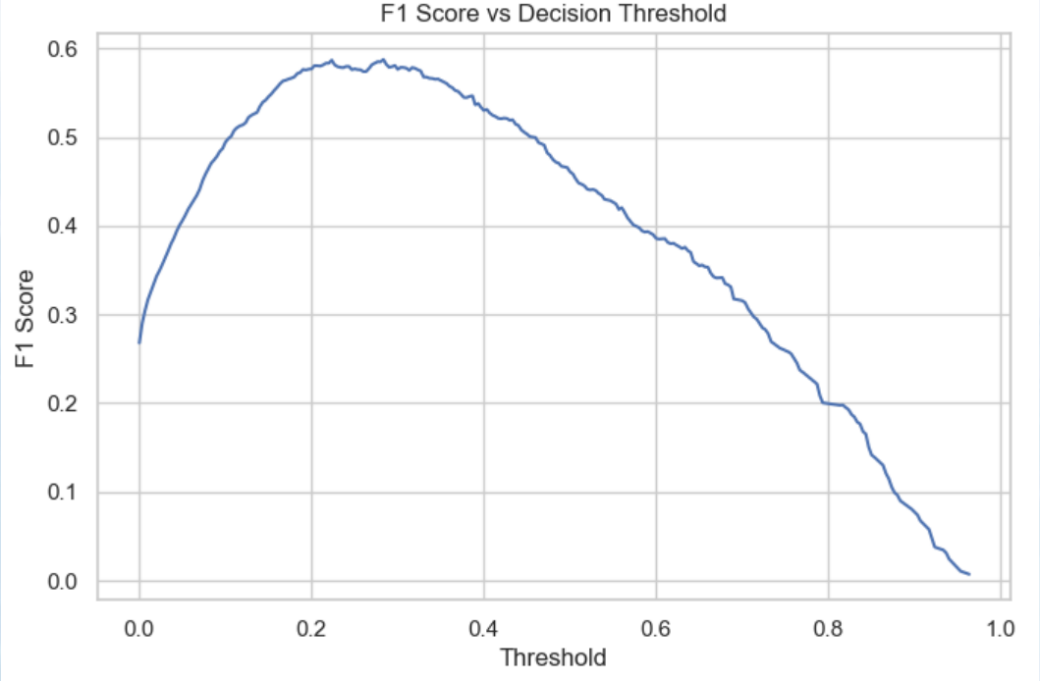
Autoencoder Findings

- Stable convergence
- Latent dimension = 12
- Reconstruction error preserved class separation
- Latent-only classifier ≠ outperform ensemble

LIVERPOOL JOHN MOORES UNIVERSITY
upGrad

Hybrid & Calibration



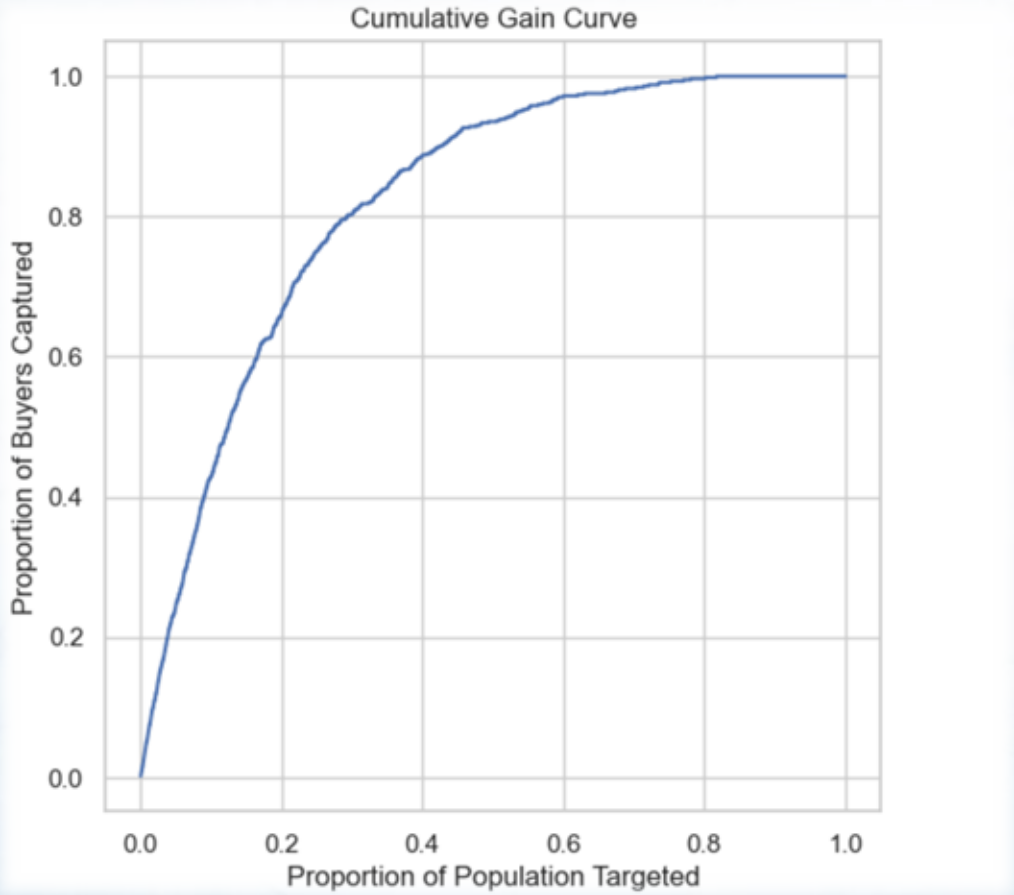


Cumulative Gain Analysis

- Top decile captures majority of buyers
- Ranking-based targeting effective

LIVERPOOL JOHN MOORES UNIVERSITY
upGrad

Business Implication



Autoencoder Findings

- Stable convergence
- Latent dimension = 12
- Reconstruction error preserved class separation
- Latent-only classifier ≠ outperform ensemble

Key Contributions

Empirical comparison of engineered behavioural features and latent representations

- Robust evaluation using multi-seed cross-validation
- Integrated analysis of calibration and threshold optimisation
- Hybrid modelling experiment combining engineered and latent features
- Behavioural interpretation of misclassification patterns

Conclusion & Future Work

LIVERPOOL JOHN MOORES UNIVERSITY

Conclusions

- Engineered behavioural features highly effective
- Ensemble modelling robust
- Latent compression informative but not superior
- Hybrid integration = limited additive value
- Stability & calibration critical

Future Directions

- Sequential modelling
- Multi-session analysis
- Cost-sensitive optimisation
- Advanced tabular transformers